17.1: Genes specify proteins via transcription and translation

Inherited traits like albinism are determined by genes, which have information content in the form of specific sequences of nucleotides along stretches of DNA, the genetic material. The DNA inherited by an organism leads to specific traits by dictating the synthesis of proteins and of RNA molecules involved in protein synthesis. In the case of coat and skin color, the gene containing the information to synthesize an enzyme that makes pigment allows normal coloring, and a faulty version of that gene leads to lack of color, or the albino phenotype. Proteins are the link between genotype and phenotype. ****Gene expression** is the process by which DNA directs the synthesis of proteins (or, in some cases, just RNAs). The expression of genes that code for proteins includes two stages: transcription and translation.

Before going into the details of how genes direct protein synthesis, let's step back and examine how the fundamental relationship between genes and proteins was discovered.

Evidence from Studying Metabolic Defects

In 1902, British physician Archibald Garrod was the first to suggest that genes dictate phenotypes through enzymes, proteins that catalyze specific chemical reactions in the cell. He postulated that the symptoms of an inherited disease reflect an inability to make a particular enzyme. He later referred to such diseases as "inborn errors of metabolism." For example, people with a disease called alkaptonuria have black urine because it contains a chemical called alkapton, which darkens upon exposure to air. Garrod reasoned that these people cannot make an enzyme that breaks down alkapton, so alkapton is expelled in their urine.

Several decades later, research supported Garrod's hypothesis that a gene dictates the production of a specific enzyme, later named the *one gene–one enzyme hypothesis*. Biochemists learned that cells synthesize and degrade most organic molecules via metabolic pathways, in which each chemical reaction in a sequence is catalyzed by a specific enzyme (see Concept 8.1). Such metabolic pathways lead, for instance, to the synthesis of the pigments that give the brown donkey in **Figure 17.1** its fur color or fruit flies (*Drosophila*) their eye color (see Figure 15.3). In the 1930s, the American biochemist and geneticist George Beadle and his French colleague Boris Ephrussi speculated that in *Drosophila*, each mutation affecting eye color blocks pigment synthesis at a specific step by preventing production of the enzyme that catalyzes that step. But neither the chemical reactions nor the enzymes that catalyze them were known at the time.

Nutritional Mutants in Neurospora: *Scientific Inquiry*

A breakthrough in demonstrating the relationship between genes and enzymes came a few years later at Stanford University, where Beadle and Edward Tatum began working with a bread mold, *Neurospora crassa*. Unlike the diploid organisms studied by Mendel (peas) and T. H. Morgan (fruit flies), *Neurospora* is a haploid species. Because its genome contains only one copy of each gene, that single copy determines the individual's expressed phenotype. Therefore, when Beadle and Tatum wanted to discover the function of any gene, they only needed to mutate and disable that one allele. They chose to study the protein-coding genes required for a specific nutritional activity. They caused mutations in genes by bombarding *Neurospora* with X-rays and looked among the survivors for mutants that differed in their nutritional needs from the wild-type bread mold.

Wild-type *Neurospora* has modest food requirements. It can grow in the laboratory on *minimal medium*—a simple solution containing minimal nutrients for growth—inorganic salts, glucose, and the vitamin biotin—incorporated into agar, a support medium. From these basics, wild-type mold cells use their metabolic pathways to produce all the other nutrient molecules (like amino acids) they need for growth, dividing repeatedly and forming visible colonies of genetically identical cells. As shown in **Figure 17.2**, Beadle and Tatum generated different "nutritional mutants" of *Neurospora* cells, each of which had a mutation in one gene and was unable to synthesize a particular essential nutrient. Such cells could not grow on minimal medium but could grow on *complete medium*, which contains all nutrients needed for growth. For *Neurospora*, the complete medium consists of the minimal medium supplemented with all 20 amino acids and a few other nutrients. Beadle and Tatum hypothesized that in each nutritional mutant, the gene for the enzyme that synthesizes a particular nutrient had been disabled by the mutation. They were able to determine experimentally which nutrient each mutant strain (original cell and its descendants) was unable to synthesize.

Figure 17.2

Beadle and Tatum's experimental approach.

Presents Beadle and Tatum's experimental approach in six steps.

To obtain nutritional mutants, Beadle and Tatum exposed Neurospora cells to X-rays, inducing mutations, then screened for (tested for) mutants that had new nutritional requirements, such as arginine, as shown here.

Beadle and Tatum's experimental approach in six steps: 1. Individual Neurospora cells were placed on complete medium. 2. The cells were subjected to X-rays to induce mutation in one gene per cell. 3. Each surviving cell formed a colony of genetically identical cells. 4. Cells from each colony were placed in a vial with only minimal medium. Cells that did not grow were identified as nutritional mutants. 5. Cells from one nutritional mutant colony were placed in a series of vials, each containing minimal medium plus one

additional nutrient. This step resulted in four vials containing the following media: minimal medium + valine, minimal medium + lysine, minimal medium + arginine, and the control wild-type cells growing on minimal medium. 6. The vials were observed for growth. In this example, the mutant cells grew only on minimal medium + arginine, indicating that this mutant was missing the enzyme for the synthesis of arginine.

This approach resulted in a valuable collection of mutant strains of *Neurospora*, each catalogued by its defect in a particular pathway. Two colleagues, Adrian Srb and Norman Horowitz, used a collection of arginine-requiring mutants to investigate the biochemical pathway for arginine synthesis in *Neurospora***(Figure 17.3)**. Srb and Horowitz pinned down each mutant's defect more specifically, using additional tests to distinguish among three classes of arginine-requiring mutants. Mutants in each class required a different set of compounds along the arginine-synthesizing pathway, which has three steps. These results, and those of many similar experiments done by Beadle and Tatum, suggested that each class was blocked at a different step in this pathway because mutants in that class lacked the enzyme that catalyzes the blocked step.

Figure 17.3

Inquiry: Do individual genes specify the enzymes that function in a biochemical pathway?

Experiment

Working with the mold *Neurospora crassa*, Adrian Srb and Norman Horowitz, then at Stanford University, used Beadle and Tatum's experimental approach (see Figure 17.2) to isolate mutants that required arginine in their growth medium. The researchers showed that these mutants fell into three classes, each defective in a different gene. From studies by others on mammalian liver cells, they suspected that the metabolic pathway of arginine biosynthesis involved a precursor nutrient and the intermediate molecules ornithine and citrulline, as shown in the diagram.

A postulated arginine synthesizing pathway. Precursor with enzyme A produces ornithine. Ornithine with enzyme B produces Citruline. Citruline with enzyme C produces Arginine.

Their most famous experiment, shown here, tested both the *one gene–one enzyme hypothesis* and their postulated arginine-synthesizing pathway. In this experiment, they grew their three classes of mutants under

the four different conditions shown in the results table below. They included minimal medium (MM) as a control, knowing that wild-type cells could grow on MM but mutant cells could not. (See test tube diagram.)

Two test tubes: (1) growth: wild type cells growing and dividing; (2) no growth: mutant cells cannot grow and divide. Both contain minimal medium. Control: minimal medium.

Results

As shown in the table, the wild-type strain was capable of growth under all experimental conditions, requiring only the minimal medium. The three classes of mutants each had a specific set of growth requirements. For example, class II mutants could not grow when ornithine alone was added but could grow when either citrulline or arginine was added.

A table illustrates the experimental results for classes of Neurospora crassa grown in certain conditions along with the summary of the results.

Two test tubes:

(1) growth: wild type cells growing and dividing;

(2) no growth: mutant cells cannot grow and divide. Both contain minimal medium.

Control: minimal medium. Flow chart. Hypothesized metabolic pathway of arginine biosynthesis: Precursor goes to Ornithine with Enzyme A; Ornithine goes to Citrulline with Enzyme B; Citrulline goes to Arginine with Enzyme C. Two tables are given. First:

Results table Classes of Neurospora crassa
Condition
Minimal medium (M M control)
M M + ornithine
M M + citrulline
M M + arginine (control)
Summary of Results:

Conclusion

From the growth requirements of the mutants, Srb and Horowitz deduced that each class of mutant was unable to carry out one step in the pathway for synthesizing arginine, presumably because it lacked the

necessary enzyme, as shown in the table. Because each of their mutants was mutated in a single gene, they concluded that each mutated gene must normally dictate the production of one enzyme. Their results supported the one gene–one enzyme hypothesis, proposed by Beadle and Tatum, and also confirmed that the arginine pathway described in the mammalian liver also operates in *Neurospora*. (Notice in the results table that a mutant can grow only if supplied with a compound made *after* the defective step because this bypasses the defect.)

A table illustrates genes coding for enzymes, and the effect of mutations in genes A, B, and C on the pathway.

Data from A. M. Srb and N. H. Horowitz, The ornithine cycle in *Neurospora* and its genetic control, *Journal of Biological Chemistry* 154:129–139 (1944).

Gene (codes for enzymes)	Wild type	Class Roman numeral 1 mutants (mutation in gene A)	Class Roman numeral 2 mutants (mutation in gene B)	Class Roman numeral 3 mutants (mutation in gene C)
Gene A	Enzyme A (Precursor goes to Ornithine)	Enzyme A crossed out in the step Precursor goes to Ornithine.	Enzyme A (Precursor goes to Ornithine)	Enzyme A (Precursor goes to Ornithine)
Gene B	Enzyme B (Ornithine goes to Citrulline)	Enzyme B (Ornithine goes to Citrulline)	Enzyme B crossed out in the step Ornithine goes to Citrulline)	Enzyme B (Ornithine goes to Citrulline)
Gene C	Enzyme C (Citrulline goes to Arginine)	Enzyme C (Citrulline goes to Arginine)	Enzyme C (Citrulline goes to Arginine)	Enzyme C crossed out in the step Citrulline goes to Arginine.

WHAT IF? *Suppose the experiment had shown that class I mutants could grow only in MM supplemented by ornithine or arginine and that class II mutants could grow in MM supplemented by citrulline, ornithine, or arginine. What conclusions would the researchers have drawn from those results regarding the biochemical pathway and the defect in class I and class II mutants?*

*For suggested answer, see*Appendix A.**

Because Beadle and Tatum set up their experimental conditions so that each mutant was defective in a single gene, the collected results, taken together, provided strong support for the *one gene–one enzyme hypothesis*, as they dubbed it: that the function of a gene is to dictate the production of a specific enzyme. Further support for this hypothesis came from experiments that identified the specific enzymes lacking in the mutants. Beadle and Tatum shared a Nobel Prize in 1958 for "their discovery that genes act by regulating definite chemical events" (in the words of the Nobel committee).

Today, we know of countless examples in which a mutation in a gene causes a faulty enzyme that in turn leads to an identifiable condition. The albino donkeys in Figure 17.1 lack a key enzyme called tyrosinase in the metabolic pathway that produces melanin, a dark pigment. The absence of melanin causes white fur and other effects throughout the donkey's body. Its nose, ears, and hooves, as well as its eyes, are pink because no melanin is present to mask the reddish color of the blood vessels that run through those structures.

The Products of Gene Expression: A Developing Story

As researchers learned more about proteins, they made revisions to the one gene–one enzyme hypothesis. First of all, not all proteins are enzymes. Keratin, the structural protein of animal hair, and the hormone insulin are two examples of nonenzyme proteins. Because proteins that are not enzymes are nevertheless gene products, molecular biologists began to think in terms of one gene–one protein. However, many proteins are constructed from two or more different polypeptide chains, and each polypeptide is specified by its own gene. For example, hemoglobin—the oxygen-transporting protein of vertebrate red blood cells—contains two kinds of polypeptides, and thus two genes code for this protein, one for each type of polypeptide (see Figure 5.18). Beadle and Tatum's idea was therefore restated as the *one gene–one polypeptide hypothesis*. Even this description is not entirely accurate, though. First, in many cases, a eukaryotic gene can code for a set of closely related polypeptides via a process called alternative splicing, which you will learn about later in this chapter. Second, quite a few genes code for RNA molecules that have important functions in cells even though they are never translated into protein. For now, we will focus on genes that do code for polypeptides. (Note that it is common to refer to these gene products as proteins—a practice you will encounter in this book—rather than more precisely as polypeptides.)

Basic Principles of Transcription and Translation

Genes provide the instructions for making specific proteins, but a gene does not build a protein directly. The bridge between DNA and protein synthesis is the nucleic acid RNA. RNA is chemically similar to DNA except that it contains ribose instead of deoxyribose as its sugar and has the nitrogenous base uracil rather than thymine (see Figure 5.23). Thus, each nucleotide along a DNA strand has A, G, C, or T as its base, and each

nucleotide along an RNA strand has A, G, C, or U as its base. An RNA molecule usually consists of a single strand.

We often describe the flow of information from gene to protein in terms of languages. Just as specific sequences of letters communicate information in a language such as English, specific sequences of monomers convey information in polymers like nucleic acids and proteins. In DNA or RNA, the monomers are the four types of nucleotides listed previously, which differ in their nitrogenous bases. Genes are typically hundreds or thousands of nucleotides long, each gene having a specific sequence of nucleotides. Each polypeptide of a protein also has monomers arranged in a particular linear order (the protein's primary structure; see Figure 5.18), but its monomers are amino acids. Thus, nucleic acids and proteins contain information written in two different chemical languages. Getting from DNA to protein involves two major stages: transcription and translation.

Transcription is the synthesis (production) of RNA using information in the DNA. The two nucleic acids are written in different forms of the same language, and the information is simply transcribed, or "rewritten," from DNA to RNA. Just as a DNA strand provides a template for making a new complementary strand during DNA replication (see Concept 16.2), it also can serve as a template for assembling a complementary sequence of RNA nucleotides. For a protein-coding gene, the resulting RNA molecule is a faithful transcript of the gene's protein-building instructions. This type of RNA molecule is called **messenger RNA (mRNA)** because it carries a genetic message from the DNA to the protein-synthesizing machinery of the cell. (Transcription is the general term for the synthesis of *any* kind of RNA on a DNA template. Later, you will learn about some other types of RNA produced by transcription.)

Translation is the synthesis of a polypeptide using the information in the mRNA. During this stage, there is a change in language: The cell must translate the nucleotide sequence of an mRNA molecule into the amino acid sequence of a polypeptide. The sites of translation are **ribosomes**, molecular complexes that facilitate the orderly linking of amino acids into polypeptide chains.

Transcription and translation occur in all organisms. Because most studies have involved bacteria and eukaryotic cells, they are our main focus in this chapter. Our understanding of transcription and translation in archaea lags behind, but we do know that archaeal cells share some features of gene expression with bacteria and others with eukaryotes.

The basic mechanics of transcription and translation are similar for bacteria and eukaryotes, but there is an important difference in the flow of genetic information within the cells. Bacteria do not have nuclei. Therefore, nuclear membranes do not separate bacterial DNA and mRNA from ribosomes and the other protein-synthesizing equipment **(Figure 17.4a)**. This lack of compartmentalization allows translation of an mRNA to begin while its transcription is still in progress, as you'll see later. By contrast, eukaryotic cells have nuclei. The presence of a nuclear envelope separates transcription from translation in space and time

(Figure 17.4b). Transcription occurs in the nucleus, but the mRNA must be transported to the cytoplasm for translation. In eukaryotes, before RNA transcripts from protein-coding genes can leave the nucleus, they are modified in various ways to produce the final, functional mRNA. The transcription of a protein-coding eukaryotic gene results in *pre-mRNA*, and further RNA processing yields the finished mRNA. The initial RNA transcript from any gene, including those specifying RNA that is not translated into protein (such as tRNA and rRNA, which will be described in Concept 17.4), is more generally called a **primary transcript**.

Figure 17.4

```
    Overview: the roles of transcription and translation in the
flow of genetic information.
```

The roles of transcription and translation in the flow of genetic information.

In a cell, inherited information flows from DNA to RNA to protein. The two main stages of information flow are transcription and translation. A miniature version of part (a) or (b) accompanies several figures later in the chapter as an orientation diagram to help you see where a particular figure fits into the overall scheme of gene expression.

The parts are as follows.

•Part a. Bacterial cell. In a bacterial cell, which lacks a nucleus, m R N A produced by transcription is immediately translated without additional processing. In the cytoplasm, D N A is transcribed into m R N A and then information in m R N A is translated in ribosome into polypeptide. Transcription and R N A processing take place inside nucleus (with nuclear envelope labeled). Translation takes place in the cytoplasm.

•Part b. Eukaryotic cell. The nucleus provides a separate compartment for transcription. The original R N A transcript, called pre-m R N A, is processed in various ways before leaving the nucleus as m R N A. Transcription: Inside the nucleus, double-stranded D N A is transcribed into pre-m R N A (single strand). R N A Processing: Pre-m R N A is processed into m R N A. Translation: m R N A leaves the nucleus and enters the cytoplasm. It is translated in the ribosomes, where polypeptides are synthesized.

Animation

Overview of Protein Synthesis in Bacteria

image

Play

image

Play

**

Animation

Overview of Protein Synthesis in Eukaryotes

image

Play

image

Play

To summarize: Genes program protein synthesis via genetic messages in the form of messenger RNA. Put another way, cells are governed by a molecular chain of command with a directional flow of genetic information:

D N A flows to R N A flows to Protein.

This idea, that the flow of information went only one way, was named the *central dogma* by Francis Crick in 1956. In the 1970s, however, scientists were surprised to discover some enzymes that use RNA molecules as templates for DNA synthesis, an example of information flow from RNA to DNA (see Concept 19.2). Still, these exceptions do not discredit the idea that, in general, genetic information flows from DNA to RNA to protein. In the next section, we discuss how the instructions for assembling amino acids into a specific order are encoded in nucleic acids.

The Genetic Code

When biologists began to suspect that the instructions for protein synthesis were encoded in DNA, they recognized a problem: There are only four nucleotide bases to specify 20 amino acids. Thus, the genetic code cannot be a language where each written symbol corresponds to a word. How many nucleotides, then, would turn out to correspond to an amino acid?

Codons: Triplets of Nucleotides

If each kind of nucleotide base were translated into an amino acid, only four amino acids could be specified, one per nucleotide base. Would a language of two-letter code words suffice? The two-nucleotide sequence AG, for example, could specify one amino acid, and GT could specify another. Since there are four possible

nucleotide bases in each position, this would give us 16 (that is, 4×4, or 42) possible arrangements—still not enough to code for all 20 amino acids.

Triplets of nucleotide bases are the smallest units of uniform length that can code for all the amino acids. If each arrangement of three consecutive nucleotide bases specifies an amino acid, there can be 64 (that is, 43) possible code words—more than enough to specify all the amino acids. Experiments have verified that the flow of information from gene to protein is based on a **triplet code**: The genetic instructions for a polypeptide chain are written in the DNA as a series of nonoverlapping, three-nucleotide words. The series of words in a gene is transcribed into a complementary series of nonoverlapping, three-nucleotide words in mRNA, which is then translated into a chain of amino acids **(Figure 17.5)**.

Figure 17.5

```
The triplet code.
```

Ribbon model of D N A with three regions labeled: Genes 1, 2 and 3. Arrows zoom in on a portion of Gene 3 to illustrate transcription and translation from D N A template.

D N A template is: 3 prime-A C C A A A C C G A G T-5 prime. (Non template strand pairs T with A and G with C following rules of complementary base pairing). During transcription m R N A is made. Nucleotide sequence is: 5 prime-U G G U U U G G C U C A-3 prime. During translation, each three-nucleotide (triplet) sequence is read as codon in the synthesis of a polypeptide (chain of amino acids). Codons from m R N A are translated into amino acids as follows:.

U G G = T r p

U U U = P h e

G G C = G l y

U C A = S e r

The protein shows the amino acids added during translation linked in a chain that reads: T r p, P h e, G l y, S e r.

For each gene, one DNA strand functions as a template for transcription of RNAs, such as mRNA. The base-pairing rules for DNA synthesis also guide transcription, except that RNA is made with uracil (U) instead of thymine (T). During translation, the mRNA is read as a sequence of nucleotide triplets, called codons. Each codon specifies an amino acid to be added to the growing polypeptide chain. The mRNA is read in the 5′→3′ direction.

VISUAL SKILLS *By convention, the nontemplate strand, also called the coding strand, is used to represent a DNA sequence. Write the sequence of the mRNA strand and the nontemplate strand—in both cases reading from* 5′ *to* 3′ *—and compare them. Why do you think this convention was adopted? (Hint: Why is this called the coding strand?)*

*For suggested answer, see*Appendix A*.*

MP3 Tutor

DNA to RNA to Protein

Hey there and welcome to another MP3 Tutor session. This is Eric Simon and I'll guide you through today's topic: the flow of biological information from DNA to RNA to proteins. This topic is arguably one of the most important in all of biology. In fact, Francis Crick, one of the co-discoverers of the structure of DNA, referred to this idea as the "Central Dogma" of biology, a tongue-in-cheek reference to the importance of understanding how genetic information is transmitted. Of course, when biologists use the term "dogma," we do not use it in the religious sense of an unchanging truth. Instead, we use it simply to highlight the importance of these ideas. In science, even the Central Dogma has been modified by further discoveries. You'll see what I mean in a minute. DNA-deoxyribonucleic acid-is an information storage molecule. Stored in its sequence of bases is all the information that was necessary to produce and maintain you as an individual. Your genetic makeup is encoded in your DNA. Your physical makeup arose through interactions of the various proteins coded by your DNA with the environment. The Central Dogma traces the flow of genetic information from DNA to proteins. Let's explore this idea in more detail and, while we do, create a diagram that can serve as a framework to organize your understanding of this important topic. At this point it would be good to gather a piece of paper and something to write with, multiple colors if possible. Taking the time to commit these ideas to paper will enhance your learning. But don't worry if you don't have paper and pen handy, you'll still be able to follow along. As the genetic material, DNA has two very important functions. First, it has to be able to direct the copying of itself, since one function of the genetic material is to pass information from one generation to the next during reproduction. The second function of DNA is to direct the cellular processes that are essential to a living organism. Most of these processes are controlled and directed by proteins, the worker bees of your body. The information stored in the genetic material directs the construction of thousands of different kinds of proteins. Crick's original Central Dogma can be stated very simply: DNA makes RNA makes proteins. The information to build a protein is stored in DNA. That information is transferred to a strand of mRNA, an intermediate information molecule that serves as a template for the building of a specific protein. In other words, the flow of information follows a path from DNA to messenger RNA to protein. As we'll see in a minute, the flow of information can follow this simple path, or it may follow a more complex one. To help you visualize the process, let's diagram this flow of

genetic information. On a standard piece of paper, write a title for this diagram at the top, perhaps something like "Central Dogma and Beyond." Now let's write some large titles down the center of the page. About one-quarter of the way down, write "DNA" in large print. Halfway down, directly under the DNA, write "mRNA." The "m" is usually lowercase and "RNA" uppercase. Below mRNA, towards the bottom of your page, write "Protein." Connect DNA to mRNA to protein with thick arrows to indicate the flow of genetic information. Make sure the arrows point from DNA to mRNA and from mRNA to Protein. This diagram represents the production of proteins via the steps of the Central Dogma. But remember that DNA performs another vital function: self-reproduction. To indicate that DNA can produce a copy of itself, draw an arrow that leaves out the right side of "DNA," goes up and around and circles back counterclockwise to point to the left side of the "DNA" label. Each of the arrows you've drawn represents an important cellular process, so let's label them. Use a different color ink if you have it. When DNA produces a copy of itself the process is called DNA replication. So label the circular arrow "Replication." You could put the words "DNA polymerase" in parentheses to remind yourself that this is the molecule that performs DNA replication. Next, label the arrow that connects DNA to mRNA as "Transcription." What does that word mean? Well, you are listening to this tutor session as an audio file. We also offer this tutorial in printed form so that you can follow along with the script. To convert my spoken words to printed words involves a process known as transcription. For instance, a court reporter transcribes the spoken proceedings of a trial into the written court record. The key point here is that the message and language is the same before and after the transcription. A similar process occurs when a small segment of DNA (a gene) is transcribed into mRNA. The language and message is still the same-the language of nucleic acids and their sequence of bases-it's just a different form of that language, from DNA to RNA. Thus, the information in DNA is copied into a molecule of mRNA through the process of transcription. You could put the words "RNA polymerase" in parentheses to remind yourself that this is the molecule that performs transcription. The language of proteins is coded in the sequence of amino acids-this is a different "language" than the language of DNA. There are only four bases in nucleic acids while most proteins are composed of 20 different amino acids. The string of nucleic acid bases is "read" in triplets called codons. A codon is a sequence of three bases that represents one amino acid, except for the three codons that serve as "Stop" signals. So, out of the 64 possible codons, 61 code for amino acids. How is the language of nucleotide triplets translated into the language of amino acids? A molecule called tRNA, or transfer RNA, performs this translation. One end of the tRNA molecule contains an anti-codon that pairs to a particular mRNA codon, and the other end of the tRNA molecule is an amino acid binding site that binds only one kind of amino acid. A molecule of tRNA thus speaks the nucleotide language on one end and the amino acid language on its other end. tRNA can therefore serve as a translator. Label the arrow that travels from mRNA to Protein as "Translation." You could put "tRNAs" in parentheses to remind yourself that these are the molecules that perform translation. This completes the first phase of your diagram. It effectively summarizes the Central Dogma as originally devised and will help you keep the entire process in mind as you study. Your job is to modify this diagram as you learn more about molecular biology.

The Central Dogma has been greatly modified over the years as our knowledge of molecular biology has increased. Here are some examples of extensions of the Central Dogma. 1 Some viruses, called retroviruses, such as HIV, store their genetic information as RNA. To be replicated, retroviruses must first convert their RNA genomes into DNA. This process, RNA to DNA, is the reverse of transcription. Draw an arrow moving from RNA to DNA in your diagram and label it "reverse transcription" with "retroviruses" in parentheses. 2 Genes stored in the DNA of eukaryotic organisms often have to be processed. (You can read about exons and introns in your text.) Many genes are therefore capable of producing multiple proteins, depending on how they are processed. In your diagram, you might want to add several arrows between DNA and mRNA and label this as "mRNA processing." 3 Gene expression is highly regulated. Genes are turned on and turned off depending on the job of a particular cell, or the time or developmental state. You could draw a box alongside the transcription arrow and label it "Gene regulation" to remind yourself of this fact. 4 Some DNA genes produce RNAs other than mRNAs. For example, there are genes for the tRNAs used in translation and for rRNAs, which are components of ribosomes. You could draw additional transcription arrows from DNA off to the sides toward boxes labeled tRNA and rRNA. What I just described are only some of the modifications to the Central Dogma discovered during the 50 years of research that have taken place since Crick first proposed the idea. The Central Dogma-DNA makes RNA makes protein-still serves as an excellent organizer for your learning. Use your diagram and modify it to organize your understanding as you explore DNA and molecular biology. I hope this has been helpful. If you need more help, please refer to your book, lecture notes, CD-ROM, or the book's website. Good luck with your studies and I'll see you on another MP3 tutorial.

During transcription, the gene determines the sequence of nucleotide bases along the length of the RNA molecule that is being synthesized. For each gene, only one of the two DNA strands is transcribed. This strand is called the **template strand** because it provides the pattern, or template, for the sequence of nucleotides in an RNA transcript. For any given gene, the same strand is used as the template every time that gene is transcribed. However, farther along on the same chromosomal DNA molecule, the opposite strand may function as the template for a different gene. Particular DNA sequences associated with a gene determine how the enzyme that transcribes genes is oriented when it binds, and this establishes which strand will be used as the template.

An mRNA molecule is complementary rather than identical to its DNA template because RNA nucleotides are assembled on the template according to base-pairing rules (see Figure 17.5). The pairs are similar to those that form during DNA replication, except that U (the RNA substitute for T) pairs with A and the mRNA nucleotides contain ribose instead of deoxyribose. Like a new strand of DNA, the RNA molecule is synthesized in an antiparallel direction to the template strand of DNA. (To review what is meant by “antiparallel” and the 5′ and 3′ ends of a nucleic acid chain, see Figure 16.7.) In the example in Figure 17.5, the nucleotide triplet ACC along the DNA template strand (written as 3′-ACC-5′) provides a template for 5′-UGG-3′ in the mRNA molecule. The mRNA nucleotide triplets are called **codons**, and they are customarily

written in the 5′→3′ direction. In our example, UGG is the codon for the amino acid tryptophan (abbreviated Trp, or W). The term *codon* is also used for the DNA nucleotide triplets along the *nontemplate* strand. These codons are complementary to the template strand and thus identical in sequence to the mRNA (except for T's instead of U's). For this reason, the nontemplate DNA strand is often called the **coding strand**; by convention, the sequence of the coding strand is used when a gene's sequence is reported.

During translation, the sequence of codons along an mRNA molecule is decoded, or translated, into a sequence of amino acids making up a polypeptide chain. The codons are read by the translation machinery in the 5′→3′ direction along the mRNA. Each codon specifies which one of the 20 amino acids will be incorporated at the corresponding position along a polypeptide. Because codons are nucleotide triplets, the number of nucleotides making up a genetic message must be three times the number of amino acids in the protein product. For example, 300 nucleotides along an mRNA strand code for 100 amino acids in the polypeptide it encodes.

Cracking the Code

Molecular biologists cracked the genetic code of life in the early 1960s when a series of elegant experiments disclosed the amino acid translations of each of the mRNA codons. The first codon was deciphered in 1961 by Marshall Nirenberg, of the National Institutes of Health, and colleagues. They synthesized an artificial mRNA made up of only uracil-bearing nucleotides, thus containing only one codon (UUU) over and over. When added to a test-tube mixture that contained all 20 amino acids, ribosomes, and the other components required for protein synthesis, the "poly-U" mRNA was translated into a polypeptide containing many units of the phenylalanine (Phe, or F) amino acids, strung together as a polyphenylalanine chain. Thus, Nirenberg determined that the mRNA codon UUU specifies the amino acid phenylalanine. Soon, the amino acids specified by the codons AAA, GGG, and CCC were similarly identified.

Although more elaborate techniques were required to decode mixed triplets such as AUA and CGA, all 64 codons were deciphered by the mid-1960s. As **Figure 17.6** shows, 61 of the 64 triplets code for amino acids. The other three codons act as "stop" signals, or termination codons, marking the end of translation. Notice that the codon AUG has a dual function: It codes for the amino acid methionine (Met, or M) and also functions as a "start" signal, or initiation codon. Genetic messages usually begin with the mRNA codon AUG, which signals the protein-synthesizing machinery to begin translating the mRNA at that location. (Because AUG also stands for methionine, polypeptide chains all begin with methionine when they are synthesized. However, an enzyme may subsequently remove this starter amino acid from the chain.)

Figure 17.6

```
The codon table for mRNA.
```

Codon table.

The table is as follows.

First m R N A base (5 prime end of codon)	Second m R N A base	Second m R N A base	Second m R N A base	Second m R N A base	Third m R N A base (3 prime end of codon)
blank	U	C	A	G	blank
U	U U U,	U C U,	U A U,	U G U,	U
U U C, P h e (F).	U C C,	U A C, T y r (Y).	U G C, C y s (C).	C	
U U A,	U C A,	U A A, Stop (highlighted in orange)	U G A, Stop (highlighted in orange)	A	
U U G, L e u (L)	U C G, S e r (S).	U A G, Stop (highlighted in orange).	U G G, T r p (W).	G	
C	C U U,	C C U,	C A U,	C G U,	U
C U C,	C C C,	C A C, H i s (H).	C G C,	C	
C U A,	C C A,	C A A,	C G A,	A	
C U G, L e u (L).	C C G, P r o (P).	C A G, G l n (Q).	C G G, A r g (R).	G	
A	A U U,	A C U.	A A U,	A G U,	U
A U C.	A C C,	A A C, A s n (N).	A G C, S e r (S).	C	
A U A, Î l e (I).	A C A,	A A A,	A G A,	A	
A U G, M e t (M) or start (highlighted in green)	A C G, T h r (T).	A A G, L y s (K).	A G G, A r g (R).	G	

G	G U U,	G C U,	G A U,	G G U	U
G U C.	G C C	G A C, A s p (D).	G G C,	C	
G U A,	G C A.	G A A.	G G A,	A	
G U G, V a l (V).	G C G, A l a (A).	G A G, G l u (E).	G G G, G l y (G).	G	

The three nucleotide bases of an mRNA codon are designated here as the first, second, and third bases, reading in the 5′→3′ direction along the mRNA. The codon AUG not only stands for the amino acid methionine (Met, or M) but also functions as a “start” signal for ribosomes to begin translating the mRNA at that point. Three of the 64 codons function as “stop” signals, marking where ribosomes end translation. Both one- and three-letter codes are shown for the amino acids; see Figure 5.14 for their full names.

VISUAL SKILLS *A segment in the middle of an mRNA has the sequence*5′-AGAGAACCGCGA-3′*. Using the codon table, translate this sequence, assuming the first three nucleotides are a codon.*

*For suggested answer, see*Appendix A*.*

Animation

Translation: The Genetic Code

image

Play

image

Play

Notice in Figure 17.6 that there is redundancy in the genetic code, but no ambiguity. For example, although codons GAA and GAG both specify glutamic acid (redundancy), neither of them ever specifies any other amino acid (no ambiguity). The redundancy in the code is not altogether random. In many cases, codons that are synonyms for a particular amino acid differ only in the third nucleotide base of the triplet. We will see why this is later in the chapter.

Our ability to extract the intended message from a written language depends on reading the symbols in the correct groupings—that is, in the correct **reading frame**. Consider this statement: “The red dog ate the bug.” Group the letters incorrectly by starting at the wrong point, and the result will probably be gibberish: for example, “her edd oga tet heb ug.” The reading frame is also important in the molecular language of cells. The short stretch of polypeptide shown in Figure 17.5, for instance, will be made correctly only if the mRNA

nucleotides are read from left to right 5′→3′ in the groups of three shown in the figure: UGG UUU GGC UCA. Although a genetic message is written with no spaces between the codons, the cell's protein-synthesizing machinery reads the message as a series of nonoverlapping three-letter words. The message is *not* read as a series of overlapping words—UGGUUU, and so on—which would convey a very different message.

Evolution of the Genetic Code

EVOLUTION The genetic code is nearly universal, shared by organisms from the simplest bacteria to the most complex plants and animals. The mRNA codon CCG, for instance, is translated as the amino acid proline in all organisms whose genetic code has been examined. In laboratory experiments, genes can be transcribed and translated after being transplanted from one species to another, sometimes with quite striking results, as shown in **Figure 17.7**. Bacteria can be programmed by the insertion of human genes to synthesize certain human proteins for medical use, such as insulin. Such applications have produced many exciting developments in the area of biotechnology (see Concept 20.4).

Figure 17.7

Evidence for evolution: expression of genes from different species.

Expression of genes from different species.

Because diverse forms of life share a common genetic code due to their shared ancestry, one species can be programmed to produce proteins characteristic of a second species by introducing DNA from the second species into the first.

Part a. Tobacco plant expressing a firefly gene. The yellow glow is produced by a chemical reaction catalyzed by the protein product of the firefly gene.

Part b. Mosquito larva expressing a jellyfish gene. The gene codes for green fluorescent protein (G F P) and is inserted into organisms as a reporter gene so researchers can tell if a gene of interest is being expressed.

The evolutionary significance of the code's near universality is clear. A language shared by all living things must have been operating very early in the history of life—early enough to be present in the common ancestor of all present-day organisms. A shared genetic vocabulary is a reminder of the kinship of all life.